Vanshika Shah

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INDUSTRY EXPERIENCE

***Data Analytics Senior*, Deloitte US-India,** Hyderabad, TG, India *Aug 2021 - Jun 2024*

- Led large-scale financial analytics engagements across 30+ enterprise clients (Dow, Honeywell, Rockwell Automation), analyzing datasets ranging from thousands to hundreds of millions of records, applying anomaly detection to identify duplicate journal entries, vendor/customer master-data inconsistencies, and high-risk transactions, enabling targeted risk assessments and accelerating audit timelines by over 2 weeks.
- Implemented a supervised regression model to detect anomalies in fixed-asset valuation and depreciation trends, uncovering multi-million-dollar variances driven by intercompany transfers, preventing potential financial misstatements, avoiding unnecessary investigations for the client, and reducing manual effort by 80%.
- Engineered automated data pipelines in Python and SAS to ingest, clean, and standardize multi-entity trial balance data across global reporting periods, cutting data-prep time by 70% and enabling downstream statistical modeling and risk-scoring workflows.
- Applied statistical analysis and predictive modeling in R to employee demographic and retirement-plan data across 20+ client organizations, surfacing insights that directly informed 401(k) plan design and workforce-planning decisions.
- Performed exploratory data analysis across accounts receivable, inventory, revenue, and expense datasets using SQL and SAS, detecting AR matching exceptions and slow-moving inventory patterns through feature engineering (threshold- and variance-based), informing process optimization, inventory planning, and financial risk management decisions.

***Data Analytics Intern*, Deloitte US-India,** Hyderabad, TG, India *Dec 2020 - Feb 2021*

- Leveraged SAS, SQL, and Excel to analyze and transform financial datasets containing up to 300M records, applying anomaly detection, variance analysis, and data validation techniques to uncover business-unit reporting discrepancies, currency inconsistencies, and cost-center exceptions that informed downstream investigations and risk assessments.

RESEARCH EXPERIENCE

Graduate Researcher, Bio-MIBLab, Georgia Institute of Technology, Atlanta, GA, USA (Supervisor: Dr. May Wang) *Fall 2025*

- Built a hybrid NLP clinical note summarization pipeline combining transformer-based LLMs (FLAN-T5 and a domain-adapted medical LLM) with an uncertainty-aware filtering layer using Monte Carlo dropout and NLI-based factuality checks to remove unsupported content and improving semantic faithfulness (BERTScore F1: 0.82 vs. 0.63 - 0.69 baselines) while reducing summary length by ~40%.

***Research Intern*, Georgia Institute of Technology,** Atlanta, GA, USA (Supervisor: Dr. Omer Inan) *Summer 2019*

- Built biomechanical models in OpenSim to simulate knee joint loading under varying gait conditions, leveraging regression and optimization techniques to analyze physiological responses and generate insights for osteoarthritis prevention research.

ACADEMIC PROJECTS

Efficient Hybrid Search for Music Discovery, Georgia Institute of Technology, GA, USA *Fall 2025*

- Built a FAISS-based hybrid vector search system using approximate nearest neighbor (ANN) search to fuse semantic similarity over 170K+ track embeddings with structured metadata filtering, applying k-means clustering for IVF indexing and predicate pushdown to prune non-matching clusters before distance computation.
- Designed a two-level filtering pipeline with adaptive oversampling, achieving 88% recall at 7.9ms latency, outperforming post-filtering by 4x recall in medium-selectivity conditions while matching its latency.

Hybrid Predictive-Causal Framework for Mental Health Outcomes, Georgia Institute of Technology, GA, USA *Fall 2025*
IEEE BHI 2025 Data Competition Top 3 Finish

- Built a predictive-causal framework to forecast depression severity (BDI-II) post mindfulness therapy using demographic, clinical, and behavioral data, combining Gaussian Copula augmentation with a shallow MLP to achieve $R^2 = 0.72$, outperforming CatBoost, Random Forest, and XGBoost baselines.
- Applied SHAP-based interpretability and propensity score causal inference to identify key outcome drivers, uncover subgroup-specific treatment effects, and estimate optimal therapy dosage recommendations.

EDUCATION

Georgia Institute of Technology, Atlanta, GA, USA
Master of Science in Computational Science and Engineering (GPA: 4.0/4.0) *Aug 2024 - May 2026*
Graduate Teaching Assistant for Machine Learning and Computing for Data Analysis

Vellore Institute of Technology, Chennai, TN, India
Bachelor of Technology in Electronics and Communication Engineering (GPA: 9.31/10.0) | Rank 10 out of 233 *Jul 2017 - Jun 2021*

TECHNICAL SKILLS

- **Languages & Databases:** Python, SQL, R, SAS, MATLAB
- **Machine Learning & AI:** Scikit-learn, PyTorch, TensorFlow, Keras, Statsmodels, Feature Engineering, Causal Inference, LLMs, Embeddings, Similarity Search, RAG, Statistical Inference, Predictive Modeling, Hypothesis Testing
- **Analytics & Visualization:** Pandas, NumPy, SciPy, Tableau, Power BI, Plotly, Matplotlib, Seaborn, Excel, Trifacta, ETL Pipeline